

1. What is Python and why is it called an interpreted language?

Python is a high-level, general purpose programming language used for building applications, data analysis, automation, AI, machine learning, web development etc. It is called interpreted language because Python code is executed line-by-line using an interpreter, then being converted to machine code all at once like compiled language.

2. What are the key features of Python that make it popular for beginners and professionals?

- * Simple and easy to learn
- * Interpreted language
- * Dynamically typed
- * Large standard library.
- * Support multiple programming paradigms.
- * Cross-platform
- * Huge community support.

3. What are Python's applications in real-world projects?

- * Web development
- * Data science & machine learning
- * Automation and scripting.
- * Artificial intelligence & deep learning.
- * Game development.
- * Cybersecurity and ethical hacking.
- * Embedded systems and IoT.

4. What is the difference between Python 2 and Python 3?

Python 2

- * print is a statement $\rightarrow$ print "Hello"
- * Less unicode support
- * Slower and outdated

Python 3

- * print is a function $\rightarrow$ print("Hello")
- * Full unicode support
- * Faster and used for all new development.

5. What is PEP 8 and why is it important in Python programming?

PEP 8 is the Python Enhancement proposal that provides style guidelines for writing clean and readable Python code. It is important because it:

- * Keep code consistent
- * Make code easier to read and maintain.
- * Helps team collaborate more effectively.

6. Who developed Python and in which year was it released?

Python was developed by Guido van Rossum and first released in 1991.

7. What do you mean by "dynamically typed" in Python?

It means that:

- * You do not need to specify the data type of a variable when declaring it.
- * The type is determined at runtime based on the value assigned.

Eg:- $x = 10$ # x is an integer

$x = "Hello"$ # x is a string.

8. What is the difference between a compiler and an interpreter and which does Python use?

Compiler

- * Convert entire code into machine code at once
- * Faster execution
- * Used in C, C++

Interpreter

- * Translate & execute line-by-line
- * Slower execution
- * Used in Python, JS